23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : **D3**
Team Members :

Roll No	Student Name
AM.SC.U4CSE25302	Anandanarayan P
AM.SC.U4CSE25312	Devanandana C
AM.SC.U4CSE25313	Devang S R
AM.SC.U4CSE25327	Kasinath K S

Project Title: Hotel Management System

Abstract:

This project presents a comprehensive Hotel Management System designed to streamline the execution of services provided to guests. The primary objective is to enhance operational efficiency and improve the overall customer experience by digitizing manual hotel processes. At the core of this system is a robust relational database developed using PostgreSQL, which is highly capable of managing, storing, and retrieving various types of real-time data critical to modern hotel operations.

The database efficiently handles complex information, including detailed guest profiles, secure ID proof verification records, dynamic billing statements, reservation histories, and ongoing service requests. The database schema is carefully structured based on comprehensive Entity-Relationship and UML class models to ensure optimal data normalization. By centralizing this information, the system ensures data consistency, minimizes redundancy, and maintains the ACID properties essential for transactional integrity.

The system is designed for multiple user roles, each with specific access privileges:

- **Guests:** The platform provides an intuitive user interface (UI) allowing guests to independently browse different types of available rooms and check their real-time availability. Guests can securely reserve rooms through the UI, which captures the necessary data and seamlessly transmits it to the backend database for processing and confirmation.

- **Receptionists:** The system acts as a centralized dashboard to manage front-desk operations. Receptionists can handle walk-in bookings, verify existing reservations, and seamlessly allocate room keys upon check-in. Furthermore, they can initiate and process specialized service requests—such as food delivery or room service—routing them to the respective departments. Every request is logged into a comprehensive service history within the database, which automatically updates the final guest invoice.
- **Administrators:** The system incorporates a managerial role, granting authorized personnel the ability to oversee daily operations, monitor revenue, update room pricing, and manage staff records.

To ensure data security, the application implements strict role-based access control. Ultimately, this PostgreSQL-driven solution bridges the gap between front-end user interactions and back-end data management, providing a scalable, efficient, and reliable framework that addresses the core administrative and operational challenges faced by the hospitality industry.